

AI Drug Repurposing Research Report

Research Query

Aspirin Parkinson's Disease

Generated on 10/5/2026, 5:15:43 pm

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Executive Summary

Parkinson's Disease (PD) represents a progressively debilitating neurodegenerative disorder characterized by the insidious loss of dopaminergic neurons in the substantia nigra, leading to profound motor and non-motor deficits. Current therapeutic strategies primarily focus on symptomatic relief, leaving a critical unmet need for disease-modifying agents capable of slowing or halting neurodegeneration. Aspirin, or acetylsalicylic acid, is a widely recognized non-steroidal anti-inflammatory drug (NSAID) with established anti-inflammatory, analgesic, and antiplatelet properties. Its multifaceted pharmacological profile, particularly its capacity to modulate inflammatory pathways, positions it as a compelling candidate for repurposing in neurodegenerative contexts where neuroinflammation is a central pathological driver.

Recent preclinical investigations have illuminated a particularly intriguing avenue involving novel aspirin derivatives, specifically NOSH-aspirin (NBS-1120). This hybrid compound, designed to release both nitric oxide (NO) and hydrogen sulfide (H₂S) in addition to its cyclooxygenase (COX) inhibitory actions, has demonstrated significant neuroprotective effects in rodent models of Parkinson's disease. These findings suggest that a targeted approach with advanced aspirin formulations could offer a superior therapeutic advantage by simultaneously addressing oxidative stress, inflammation, and cellular dysfunction crucial to PD pathogenesis. The high market potential in the drug repurposing sector for such a well-understood pharmacological scaffold, coupled with the identification of related patents, underscores the commercial viability of pursuing these next-generation aspirin compounds. Despite the promising preclinical data, a substantial void exists in direct human clinical trial evidence validating either conventional aspirin or its derivatives specifically for PD. This necessitates rigorous translational research to bridge the gap between bench and bedside, ultimately evaluating the efficacy and safety of aspirin-based therapies in the Parkinsonian patient population.

Disease Background

Parkinson's Disease is a complex and relentlessly progressive neurodegenerative disorder, primarily defined by the degeneration of dopaminergic neurons within the substantia nigra pars compacta. This neuronal loss results in a severe deficit of dopamine in the striatum, precipitating the characteristic motor symptoms of bradykinesia, rigidity, resting tremor, and postural instability. Beyond motor dysfunction, PD patients frequently experience a spectrum of non-motor symptoms, including cognitive impairment, autonomic dysfunction, sleep disturbances, and mood disorders, which significantly impact their quality of life and often precede motor symptom onset. The pathological hallmark of PD is the accumulation of aggregated alpha-synuclein protein in neuronal inclusions known as Lewy bodies, which propagate throughout the central and peripheral nervous systems, contributing to widespread neuronal dysfunction and cell death.

The precise etiology of PD remains largely elusive, but a confluence of genetic predispositions and environmental factors is believed to contribute to its development. At a cellular level, mitochondrial dysfunction, heightened oxidative stress, and impaired protein degradation pathways (e.g., ubiquitin-proteasome system, autophagy-lysosomal pathway) are recognized as pivotal mechanisms driving neurodegeneration. Crucially, growing evidence highlights the significant role of chronic neuroinflammation in perpetuating neuronal damage in PD. Activated microglia and astrocytes, coupled with elevated levels of pro-inflammatory cytokines and reactive oxygen species, create a hostile microenvironment that exacerbates dopaminergic neuron loss. The intricate interplay of these pathological cascades underscores the pressing need for therapeutic interventions that can not only ameliorate symptoms but also exert neuroprotective effects by targeting these fundamental disease processes. Addressing neuroinflammation, oxidative stress, and cellular resilience offers a promising avenue for developing much-needed disease-modifying treatments for Parkinson's Disease.

Drug Overview

Aspirin, chemically known as acetylsalicylic acid, is a venerable pharmaceutical agent with a history of medicinal use spanning over a century. Synthesized in its modern form in the late 19th century, it rapidly became a cornerstone of pharmacological therapy due to its diverse beneficial properties. Aspirin belongs to the class of non-steroidal anti-inflammatory drugs (NSAIDs) and is widely available globally. Its primary indications include the relief of mild to moderate pain (analgesic), reduction of fever (antipyretic), and attenuation of inflammatory conditions such as arthritis. Furthermore, at lower doses, aspirin is extensively used for its antiplatelet effects, playing a critical role in the primary and secondary prevention of cardiovascular and cerebrovascular thrombotic events, including myocardial infarction and ischemic stroke.

Pharmacokinetically, aspirin is rapidly and extensively absorbed from the gastrointestinal tract following oral administration, with peak plasma levels typically achieved within 1 to 2 hours. It is then rapidly hydrolyzed in the gut wall and liver to its primary active metabolite, salicylic acid. Aspirin itself has a very short plasma half-life of approximately 15-20 minutes, while salicylic acid exhibits a longer half-life, which is concentration-dependent and can range from 2 to 4.5 hours at typical anti-inflammatory doses, extending to 15-30 hours at higher, toxic doses due to saturation of metabolic pathways. Both aspirin and salicylic acid are widely distributed throughout the body, binding extensively to plasma proteins, particularly albumin. Elimination occurs primarily via renal excretion, with salicylic acid undergoing conjugation reactions (e.g., with glucuronic acid and glycine) before urinary excretion. The well-established pharmacokinetic and pharmacodynamic profiles of aspirin contribute to its pervasive clinical utility, making it a highly characterized compound for potential repositioning strategies.

Mechanism of Action

The pharmacological actions of aspirin are primarily mediated through the irreversible inhibition of cyclooxygenase (COX) enzymes, specifically COX-1 and COX-2. By acetylating a serine residue in the active site of these enzymes, aspirin prevents the conversion of arachidonic acid into prostaglandins, prostacyclins, and thromboxanes. Inhibition of COX-2 underlies its anti-inflammatory and analgesic effects by reducing the production of pro-inflammatory prostaglandins. Inhibition of COX-1, particularly in platelets, leads to a reduction in thromboxane A₂ synthesis, which is responsible for platelet aggregation and vasoconstriction, thereby conferring aspirin's antiplatelet effects. However, the potential for aspirin in neurodegenerative diseases like Parkinson's Disease extends beyond these traditional mechanisms, particularly with the advent of novel hybrid molecules.

A key development in this area is the engineering of NOSH-aspirin (NBS-1120), a compound designed to leverage aspirin's core activity while simultaneously delivering nitric oxide (NO) and hydrogen sulfide (H₂S). This innovative approach targets multiple facets of PD pathophysiology. Nitric oxide, a gasotransmitter, plays diverse physiological roles, including neurotransmission modulation, vasodilation, and anti-inflammatory effects, though its dual nature means it can also contribute to oxidative stress in excess. Hydrogen sulfide, another crucial gasotransmitter, is increasingly recognized for its neuroprotective properties, acting as an endogenous antioxidant, anti-inflammatory agent, and modulator of mitochondrial function. By releasing both NO and H₂S, NOSH-aspirin offers a sophisticated mechanism to attenuate motor defects and dopaminergic neuron degeneration, as observed in preclinical models of Parkinson's disease. The synergistic action of COX inhibition, NO release, and H₂S liberation allows NOSH-aspirin to combat neuroinflammation, oxidative stress, and mitochondrial dysfunction, which are central to PD progression. This multi-target engagement represents a significant advancement over conventional aspirin, offering a more comprehensive neuroprotective strategy by addressing the "boulevard of broken nerves" where neuroinflammation transitions into neurodegeneration. The involvement of gasotransmitters in PD pathophysiology further supports the rationale for this hybrid approach, providing a targeted intervention against the complex molecular cascades driving the disease.

Literature Review

The current scientific literature, particularly within the provided dataset, highlights a nascent but compelling preclinical rationale for aspirin-based therapeutics in Parkinson's Disease, with a strong emphasis on novel hybrid formulations. A pivotal study by Firouzan and colleagues (2025) directly investigated NOSH-aspirin (NBS-1120) in a rat model of Parkinson's disease. This research demonstrated that NOSH-aspirin was effective in attenuating motor defects, a critical symptom of PD, and crucially, in preventing dopaminergic neuron degeneration. These findings are significant as they provide direct preclinical evidence of a modified aspirin's neuroprotective potential, moving beyond the traditional understanding of aspirin's anti-inflammatory role to a more complex, multi-modal mechanism involving gasotransmitters. The observed reduction in neuronal loss directly addresses a fundamental pathological feature of PD.

Further reinforcing the theoretical underpinnings of this approach, O'Connor et al. (2025) detailed the intricate role of gasotransmitters—nitric oxide, carbon monoxide, and hydrogen sulfide—in Parkinson's disease pathophysiology. Their work underscores how these endogenous gaseous signaling molecules influence neuroinflammation, oxidative stress, and mitochondrial function, all of which are central to dopaminergic neuronal demise. This context is directly relevant to NOSH-aspirin, whose mechanism explicitly involves the controlled release of nitric oxide and hydrogen sulfide, thereby leveraging these protective gasotransmitter pathways. Complementing this, Attiq et al. (2025) provided a comprehensive review on the continuum from neuroinflammation to neurodegeneration, emphasizing the critical role of inflammatory processes in initiating and perpetuating neuronal damage in the central nervous system. This review implicitly supports the therapeutic strategy of anti-inflammatory agents like aspirin in neurodegenerative conditions, positioning it as a tool to disrupt the cycle of inflammation-induced neuronal demise.

While the aforementioned studies build a strong scientific case for advanced aspirin derivatives, the broader context of antiplatelet agents in neurological disease presents

complexities. Zhan et al. (2026) reported an association between clopidogrel use and an increased prevalence of Parkinson's disease, particularly in US adults with diabetes. This finding, while not directly related to aspirin, serves as an important cautionary note. It suggests that not all antiplatelet therapies confer neurological benefits, and some may even be detrimental in specific contexts, underscoring the necessity for agent-specific and context-specific investigation rather than broad generalizations about drug classes. Critically, within the provided literature, there is a distinct absence of studies directly evaluating the efficacy of conventional, unmodified aspirin in human Parkinson's disease patients, suggesting that the current research frontier is primarily focused on elucidating the potential of enhanced aspirin formulations in preclinical models rather than immediate clinical application of the traditional compound for this specific indication.

Clinical Insights

A thorough analysis of the provided clinical trial data from ClinicalTrials.gov reveals a significant gap in direct human clinical investigations into the efficacy of aspirin for Parkinson's Disease. Specifically, no active, recruiting, or completed trials were identified that directly test either conventional aspirin or its novel derivatives, such as NOSH-aspirin, as a treatment or preventative measure for PD. This absence of direct clinical evidence is a critical observation, indicating that the concept of aspirin as a therapeutic agent for Parkinson's Disease is largely confined to the preclinical research phase, particularly with the promising results observed for modified aspirin compounds.

The closest tangential clinical trial identified, NCT06715007, focuses on "Antiplatelet Therapy and Endothelial-stabilizing Agents in Cerebral Small Vessel Diseases." While this trial involves antiplatelet therapy, a primary mechanism of aspirin, and addresses cerebrovascular health, it does not specifically target Parkinson's Disease nor does it explicitly name aspirin as the investigational drug. Its status as "RECRUITING" indicates ongoing interest in the broader application of antiplatelet agents in neurological contexts, which is an important background for aspirin's potential. However, it cannot be

interpreted as direct evidence for aspirin's utility in PD. Other identified trials, NCT01285908 for orthostatic hypotension and NCT02670161 for brain tumors, are entirely unrelated to the current query concerning aspirin and Parkinson's Disease.

The lack of direct clinical trials for aspirin in PD implies that significant translational hurdles must be overcome before these preclinical findings can impact patient care. Future clinical development would need to establish not only the safety and tolerability of aspirin or its derivatives in PD patients but also critically evaluate their potential to modify disease progression. This would involve meticulously designed trials with appropriate endpoints, considering the chronic nature of PD and the need for long-term efficacy and safety data. The transition from promising rat model data for NOSH-aspirin to human trials will require substantial investment and rigorous scientific inquiry, necessitating phase 1 studies to establish human pharmacokinetics and safety, followed by larger efficacy trials.

Safety Analysis

Aspirin, as one of the most widely used pharmaceuticals globally, possesses a well-characterized safety profile based on decades of extensive clinical use in various indications. Despite its widespread availability and general tolerability, aspirin is associated with several well-documented adverse events and contraindications that must be carefully considered, particularly in the context of potentially repurposing it for a chronic condition like Parkinson's Disease. The most prominent safety concerns are related to its gastrointestinal (GI) and antiplatelet effects.

Common adverse events include dyspepsia, nausea, heartburn, and abdominal pain. More serious GI complications can involve gastric and duodenal ulcers, erosions, and potentially life-threatening GI bleeding, especially with long-term use or at higher doses. These effects are primarily due to the inhibition of COX-1, which normally protects the gastric mucosa. Given that many Parkinson's patients experience existing GI issues

such as constipation, the addition of a GI-irritating agent could exacerbate these symptoms and impact their overall quality of life. The antiplatelet action, while beneficial in cardiovascular prevention, carries an inherent risk of increased bleeding. This includes epistaxis, bruising, and a heightened risk of hemorrhagic stroke, which must be weighed against any potential neuroprotective benefits, particularly in an elderly population often on multiple medications.

Contraindications for aspirin use include a history of hypersensitivity to salicylates or other NSAIDs, active peptic ulcer disease, bleeding disorders (e.g., hemophilia, thrombocytopenia), severe hepatic or renal impairment, and severe uncontrolled hypertension. Co-administration with anticoagulants can significantly increase the risk of bleeding. For novel hybrid aspirins like NOSH-aspirin, while preclinical data may suggest an improved GI safety profile due to the local release of NO and H₂S (which can be gastroprotective), these benefits would need rigorous validation in human clinical trials. The overall safety assessment for aspirin in PD would require a careful balance of potential neuroprotective efficacy against the well-known risks, considering the polypharmacy often observed in elderly PD patients and the potential for drug-drug interactions with existing anti-Parkinsonian medications.

Market Potential

The market potential for an effective disease-modifying therapy for Parkinson's Disease is exceptionally high, given the substantial unmet medical need and the growing global prevalence of the disorder. Currently, available treatments primarily offer symptomatic relief, with no approved therapies capable of halting or reversing the progressive neurodegeneration. This creates a significant commercial opportunity for any agent that can demonstrate neuroprotective or disease-modifying effects. The concept of drug repurposing, wherein existing drugs are investigated for new indications, adds another layer of attractiveness to aspirin. As an established drug with a known safety profile, repurposing significantly de-risks the development pathway, potentially reducing the

time and cost associated with bringing a new therapy to market. The provided market intelligence explicitly notes "High potential in repurposing market," which strongly supports this view.

While generic aspirin itself is inexpensive and lacks patent protection, the market value lies squarely in the development and commercialization of novel, patented derivatives such as NOSH-aspirin. The intelligence mentioning "Some related patents identified" is crucial, as intellectual property protection is essential for pharmaceutical companies to realize a return on investment for the substantial R&D required for clinical validation. If NOSH-aspirin, or similar compounds, can demonstrate superior efficacy and a favorable safety profile specifically in PD, they could command premium pricing, similar to other innovative therapies for neurodegenerative diseases. The target patient population is substantial and continually expanding with an aging global demographic. A neuroprotective aspirin derivative could position itself as a first-in-class or best-in-class agent, significantly differentiating itself from existing symptomatic treatments. Furthermore, the extensive global market for aspirin indicates broad manufacturing capabilities and supply chain infrastructure, which, while for the generic compound, provides a robust foundation for scaling production of derivatives if clinical success is achieved. This combination of high unmet need, repurposing advantages, and patent protection for innovative formulations paints a promising market landscape.

Challenges

The pursuit of aspirin or its derivatives as a therapeutic strategy for Parkinson's Disease faces several significant scientific, clinical, and commercial challenges that must be meticulously addressed. Scientifically, translating the promising preclinical findings, particularly those related to NOSH-aspirin in rat models, into human efficacy remains a formidable hurdle. The complex pathophysiology of PD in humans, involving intricate genetic and environmental interactions, may not be fully recapitulated in animal models. Determining the optimal dose, formulation (e.g., conventional vs. hybrid), and specific

therapeutic window to achieve neuroprotection without unacceptable side effects in humans is critical. Further elucidation of the precise molecular mechanisms by which NO and H₂S contribute to neuroprotection in a living human brain, and how these interact with aspirin's traditional COX inhibition, is also required to refine future treatment strategies.

Clinically, the most pressing challenge is the complete absence of direct human clinical trial data for aspirin in PD. This necessitates initiating the entire clinical development pathway, from early-phase safety and tolerability studies to large-scale, long-term efficacy trials. Designing such trials for a chronic, slowly progressing disease like PD presents methodological difficulties, including the selection of appropriate biomarkers for disease modification and definitive clinical endpoints. The widespread use of conventional aspirin for other indications (e.g., cardiovascular protection) could confound observational studies and make it challenging to isolate its specific effect on PD progression. Furthermore, the inherent safety risks of aspirin, particularly gastrointestinal bleeding and hemorrhagic complications, are substantial concerns in an elderly PD population often on multiple medications. These risks must be carefully managed and weighed against potential benefits, particularly in light of potential drug-drug interactions with existing PD medications or other co-morbidities.

Commercially, the lack of patent protection for generic aspirin means that any commercial success hinges entirely on the development and patenting of novel derivatives like NOSH-aspirin or innovative delivery systems. While "some related patents identified" offers a positive signal, their scope and enforceability will dictate the commercial viability. Competing against established symptomatic PD therapies and a pipeline of other investigational neuroprotective agents will require a compelling efficacy and safety profile. Additionally, gaining market access and achieving favorable reimbursement for a repurposed drug, even an enhanced derivative, can be challenging if the clinical benefits are perceived as incremental or if the drug is seen as merely an "aspirin" by payers and prescribers, despite its distinct pharmacological advantages. Overcoming these multi-faceted challenges will require substantial investment, rigorous research, and strategic commercial planning.

Conclusion

The hypothesis concerning aspirin's potential in Parkinson's Disease, while largely unproven for conventional aspirin in human clinical contexts, gains compelling scientific traction through the lens of novel, hybrid aspirin derivatives. Preclinical evidence for NOSH-aspirin (NBS-1120) strongly suggests a multifaceted neuroprotective mechanism, demonstrating the attenuation of motor deficits and preservation of dopaminergic neurons in rodent models of PD. This advanced aspirin formulation, by integrating traditional cyclooxygenase inhibition with the controlled release of nitric oxide and hydrogen sulfide, strategically targets key pathological drivers of PD, including neuroinflammation, oxidative stress, and cellular dysfunction, aligning with contemporary understanding of disease pathophysiology.

Scientifically, the rationale for exploring aspirin-based therapeutics in PD is robust, driven by the demonstrated preclinical efficacy of novel derivatives and a solid mechanistic basis involving gasotransmitters and anti-inflammatory pathways. The potential to leverage an existing pharmacological scaffold significantly de-risks early-stage drug development. However, commercially and clinically, the journey ahead is arduous. The complete absence of direct human clinical trial data for any aspirin formulation in PD represents a critical barrier. Validation through rigorous phase I, II, and III clinical studies is indispensable to translate preclinical promise into a tangible therapeutic option. While the market potential for a disease-modifying PD drug is undeniably high, particularly for patented hybrid molecules, overcoming scientific translation hurdles, navigating complex clinical development, and establishing a differentiated commercial value proposition will require substantial investment and strategic execution.

In definitive scientific and commercial terms, the current data suggest that while conventional aspirin's direct role in PD is highly speculative and lacks clinical support, the **concept** of aspirin-based polypharmacology, particularly through innovative derivatives like NOSH-aspirin, holds significant and exciting promise. These next-

generation compounds offer a sophisticated approach to combating neurodegeneration. A guarded but optimistic verdict emerges, emphasizing the need for focused and well-funded translational research to unequivocally determine the efficacy and safety of these promising aspirin derivatives in the Parkinsonian patient population, potentially offering a much-needed disease-modifying therapy.

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